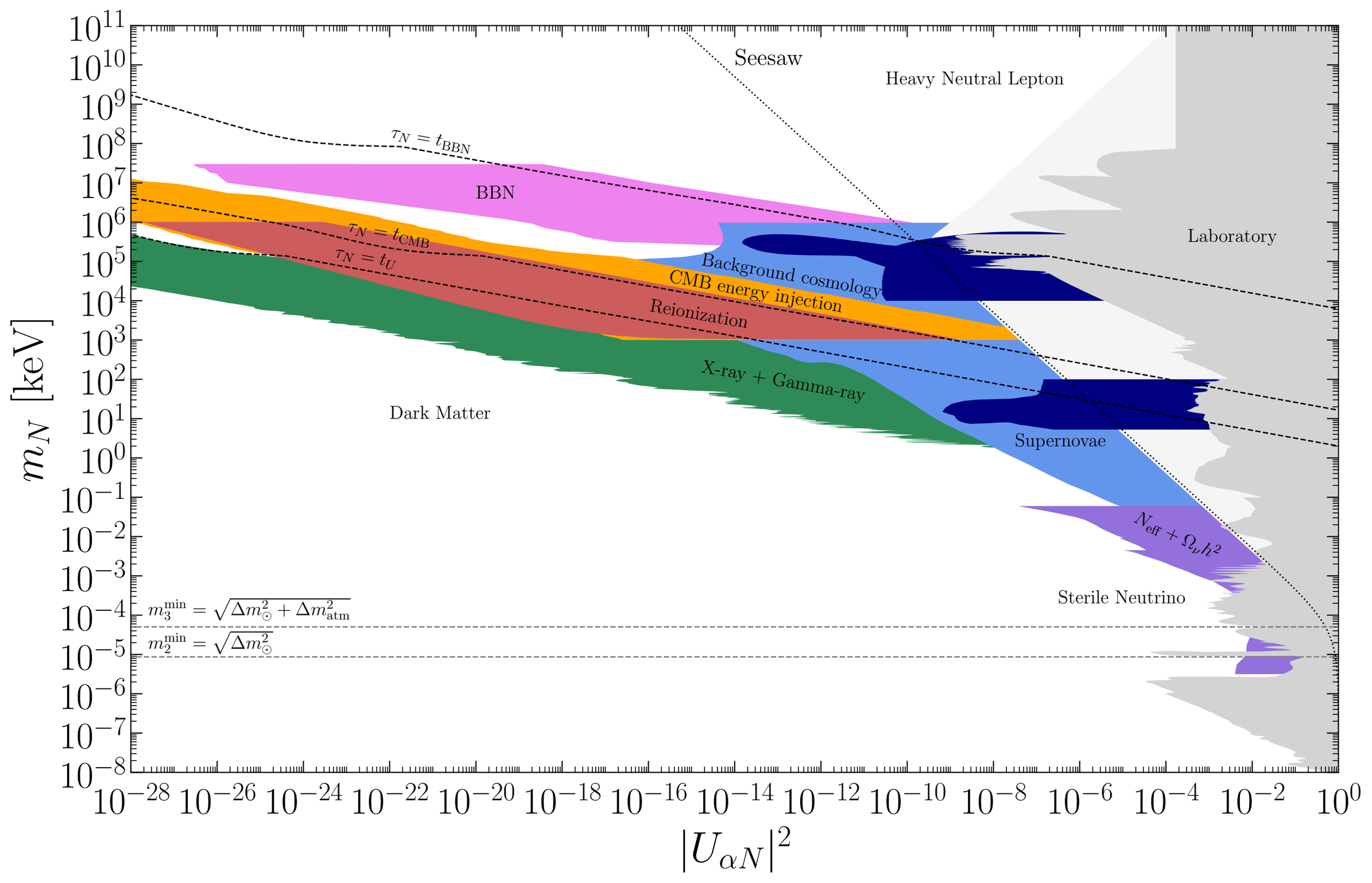

Seesaw
Heavy Neutral Lepton
$\tau_N = t_{\rm BBN}$
BBN
$\tau_N = t_{\rm CMB}$
$\tau_N = t_U$
Background cosmology
CMB energy injection
Reionization
X-ray + Gamma-ray
Dark Matter
Laboratory
Supernovae
$N_{\rm eff} + \Omega_\nu h^2$
Sterile Neutrino
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
m_N [keV]
$|U_{\alpha N}|^2$